

world to draw its own conclusions about what happens.

Lisp and Dylan have always been favourites for AI programmers because they allow the programmer to add more and more information about the world at runtime, basically, teaching the program new things.

What Is It Good For?

The goal of developing adaptive software is to make software more like living beings. Just like in living society, software agents will observe, learn from and co-operate with its environment as well as other agents. The possibilities for such software are endless.

The aim, of course, is to finally reach that Holy Grail of computing—a HAL 9000—a computer that could converse with humans as if it were a human itself. It could learn about you and speak to you only about things that you like talking about.

A HAL 9000 built without adaptability would behave the same whether it was interacting with a 40-year-old nuclear physicist or a two-year-old child. An adaptive HAL 9000, however, would discover that the interaction with two-year-old wasn't really working out so well, and would soon alter its approach to get the best results, eventually reaching the level of baby-talk.

Software will also be more resilient. Server software, running on multi-processor systems, would quickly adapt itself if one of those processors failed (providing, of course, that the hardware is built that way), keeping the server alive.

Software could be developed that would monitor your PC's performance, and give up some of its resource-hungry tasks to ensure that it is always responsive to your actions. Adaptive software would also have enough "self-awareness" to detect a malfunctioning algorithm within itself and be able to still run, albeit a little crippled.

Robots today are built for specific tasks—programming them to take on new responsibilities is cumbersome. But with adaptive software, robots can take on new roles easily. Adaptive software can also be used in robot teams, say, a robotic football team where each team member can take on a new role depending on its position on the field.

Gaming AI is at an incredible level today. Your opponents will choose fantastic strategies to get your goose, and you need to push your abilities to the limit to figure out how to beat them. But once you have figured out their shortcomings, the game becomes a snap.

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Now imagine this, a game that realises that its old strategies aren't going to work anymore. It's being defeated, and there are no rewards to being defeated, so it sets about figuring how to beat you. It observes you, gets to know you and learns your favourite moves and weaknesses. It then uses this knowledge to hit upon the best way to turn you to dust. No corner would be safe; you would have to give up your standard moves and trade them in for new ways to approach the game every time. The agents in the game would be as dangerous as real human players, perhaps even more so.

When Will This Happen?

We are already seeing little instances of adaptive software in our lives. Microsoft's Office Assistant is one of the earliest examples of software that tries to anticipate user needs. Today, Google's AdSense program has some semblance of adaptivity—it shows you advertisements it thinks you are most likely to click. Adaptability becomes more difficult to implement as the purpose gets more complex, but the Agents are already among us. You never know where one might pop up.

We should see the first examples of truly adaptive software appearing in places where the efficiency and survival of the system are foremost—server applications that can adapt to the changing load of the system, and are immune to minor hardware failures.

After this, we should see adaptive software extending itself to more complex (read: more fun) applications such as gaming. Game developers are always trying to make more challenging games; soon every computer-generated opponent will be an agent.

Perhaps, most importantly, adaptive software will take us along the road to true AI. A system can exhibit intelligent behaviour even if it's dumb—but true intelligence can only emerge if a program goes beyond what the programmer foresaw.

Intelligent software explores its surroundings, learns from them and thus evolves beyond what it was originally intended for. In other words, it should behave like a living being. Adaptive software aims precisely at that. Think about it: today's programs entirely reflect the intelligence of the programmer, who has to baby-sit the program for its lifetime. Tomorrow's programmers will be able to let their creations loose—to let them go forth, prosper and multiply! ☒

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